MS24129
Siddhant

# INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

## Mid-Term I 2025 - 2026

## IDC 214: QUANTITATIVE ESTIMATES IN SOFT MATTER

**FULL MARKS 15** **DURATION 1 HR**

1. A sphere of radius 2 mm is moving through water at 27°C. Find out its diffusion coefficient. **(3)**

2. Identical microscopic particles are diffusing in a fluid kept in a 500 mL beaker. If the RMS displacement of the particles after 10 minutes is 1. 6 mm, find out the diffusion coefficient of the particles. **(2)**

3. Thoughtful Question: Pointwise and with the help of equations, solve the problems given in the table with quantitative answers. **(5)**
**Given:**

| Bacterium (E. coli) | Human |
|---|---|
| $L \approx 2\ \mu m$ | $L \approx 1.5$ m |
| $v \approx 20\ \mu m/s$ | $v \approx 1$ m/s |
| $\rho_{water} = 1000\ kg/m^3$, $\eta_{water} = 0.001$ kg/(m.s) | Same fluid properties |

Find out the differences in a table as shown below:

| Aspect | Bacteria | Human | Comments |
|---|---|---|---|
| $R_e$ | | | |
| Drag force | | | |

4. State Purcell's Scallop Theorem and explain how it is applicable to Bacteria (you may refer to the answers from question 3). **(2)**

5. Find the terminal velocity of a spherical particle (radius 1 μm, density 1050 kg/m$^3$) falling in water ($\rho_{water}$ = 1000 kg/m$^3$). [Hint. Terminal velocity is the velocity of the particle where drag force and gravitation force are balanced.] **(3)**